

A Reduced Newtonian Model of Tidally Deformable Pseudo-Fluid Planets: Full Modal Formulation and Quadrupole Implementation

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Abstract—This paper develops a Newtonian model for tidally interacting planets whose surfaces deviate from exact sphericity through damped pseudo-fluid modes. The goal is not a full hydrodynamic interior simulation, but a structured reduced model that still contains the key mechanisms of tidal evolution: tidal bulges, gravitational backreaction of the bulges, spin torques, orbital circularization, spin synchronization, and obliquity evolution. First, a general deformable-body formulation is written in which each planet has orbital motion, rigid-body orientation, spin, and an arbitrary set of surface modes. Second, a feasible all-mode approximation is introduced using body-frame symmetric trace-free tensors and generalized rheology with relaxation branches. Third, a quadrupole truncation is derived and written in a form directly suitable for numerical integration. Compared with the simplest quadrupole tide model, the upgraded version includes shape-dependent spin inertia, rotational flattening, frequency-dependent lag through internal memory variables, and the leading quadrupole angular momentum carried by the deformation itself. This last correction restores the explicit quadrupole torque to the reduced spin equation without upsetting the conservative bookkeeping, making the model materially better suited to obliquity studies.

Index Terms—Tides, celestial mechanics, deformable bodies, spherical harmonics, quadrupole approximation, spin-orbit coupling, rheology.

I. INTRODUCTION

Many problems in celestial mechanics lie between two standard idealizations. At one extreme, planets are treated as point masses and evolve only through Newtonian N -body gravity. At the other extreme, one solves the full equations of fluid or viscoelastic motion in self-gravitating rotating bodies, including stratification, realistic rheology, and thermal physics. The first approach is simple but cannot represent tides. The second is physically rich but expensive and often problem-specific.

The purpose of this work is to formulate a middle-ground model with the following properties:

- 1) full three-dimensional Newtonian orbital dynamics,
- 2) explicit deformation degrees of freedom,
- 3) deformation-dependent gravity,
- 4) spin and orientation dynamics,
- 5) dissipative tidal lag,
- 6) a systematic path from a general modal model to a compact implementation model.

The resulting framework is still a reduced model—much closer to a “structured toy model” than to a full planetary

interior simulation. It is not an arbitrary toy, however: the couplings between orbit, spin, and deformation have the correct geometric form, and increasingly refined approximations can be introduced without changing the overall architecture.

Four specific improvements distinguish the present version from the simplest damped-quadrupole model. The spin inertia depends on the tidal deformation. Rotational flattening is handled through the same shape-dependent coefficient that enters the spin kinetic energy, so the rotational forcing and the energy bookkeeping share a single common parameter. The dissipative response supports one or more relaxation branches rather than a single constant time-lag term, allowing frequency-dependent rheology while remaining in ordinary differential equation (ODE) form. Finally, the quadrupole closure retains the leading angular momentum carried by the quadrupole mode itself; this is the minimal correction that allows the explicit quadrupole torque to appear in the spin equation without spoiling conservative bookkeeping, and it is especially important for obliquity evolution.

II. KINEMATICS AND VARIABLES

Consider N planets indexed by $a = 1, \dots, N$. For each body we introduce

- mass M_a ,
- reference radius R_a ,
- center-of-mass position $X_a(t) \in \mathbb{R}^3$,
- orientation matrix $Q_a(t) \in SO(3)$,
- body-frame angular velocity $\Omega_a(t) \in \mathbb{R}^3$.

The orientation evolves according to

$$\dot{Q}_a = Q_a \hat{\Omega}_a, \quad (1)$$

where $\hat{\Omega}_a$ is the skew-symmetric matrix satisfying

$$\hat{\Omega}_a v = \Omega_a \times v. \quad (2)$$

A point y in the body frame corresponds to inertial position

$$x = X_a + Q_a y. \quad (3)$$

This explicit separation between inertial and body frames is essential. It avoids fictitious torques in the reduced equations and makes it possible to describe tides by body-frame variables while still computing forces and torques in the inertial frame.

III. SURFACE DESCRIPTION AND EXACT DEFORMABLE BODY

A. Surface shape

Assume the body remains star-shaped with respect to its center. Its surface in the body frame is written as

$$r = \mathcal{R}_a(\hat{y}, t) = R_a(1 + u_a(\hat{y}, t)), \quad (4)$$

where $\hat{y} \in S^2$ and u_a is the dimensionless shape field.

Expand the shape in spherical harmonics:

$$u_a(\hat{y}, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} q_{a\ell m}(t) Y_{\ell m}(\hat{y}). \quad (5)$$

The occupied body-frame volume is

$$V_a(t) = \{y \in \mathbb{R}^3 : 0 \leq |y| < \mathcal{R}_a(\hat{y}, t)\}. \quad (6)$$

B. Constraints

For an incompressible pseudo-fluid, the volume is fixed:

$$\frac{1}{3} \int_{S^2} \mathcal{R}_a(\hat{y}, t)^3 d\Omega = \frac{4\pi}{3} R_a^3. \quad (7)$$

The body-frame center of mass is fixed at the origin:

$$\int_{V_a(t)} y d^3y = 0. \quad (8)$$

At linear order, these conditions eliminate independent $\ell = 0$ and $\ell = 1$ modes. The tidal content therefore begins at $\ell = 2$.

C. Exact gravity

Assume uniform density inside the deformed body:

$$\rho_a(t) = \frac{M_a}{\text{Vol}(V_a(t))}. \quad (9)$$

The exact Newtonian potential of body a is

$$\Phi_a(x, t) = -G\rho_a \int_{V_a(t)} \frac{d^3y}{|x - X_a - Q_a y|}. \quad (10)$$

For x outside the body, this admits a multipole expansion. Let

$$s = Q_a^\top (x - X_a). \quad (11)$$

Then

$$\Phi_a(x, t) = -G \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \frac{Q_{a\ell m}(t)}{|s|^{\ell+1}} Y_{\ell m}(\hat{s}), \quad (12)$$

where

$$Q_{a\ell m}(t) = \int_{V_a(t)} \rho_a r^\ell Y_{\ell m}^*(\hat{y}) d^3y. \quad (13)$$

Equivalently,

$$Q_{a\ell m}(t) = \frac{\rho_a}{\ell+3} \int_{S^2} \mathcal{R}_a(\hat{y}, t)^{\ell+3} Y_{\ell m}^*(\hat{y}) d\Omega. \quad (14)$$

D. Exact pseudo-fluid motion

Let the internal motion be irrotational in the body frame:

$$w_a(y, t) = \nabla_y \phi_a(y, t), \quad \nabla_y^2 \phi_a = 0 \quad \text{in } V_a(t). \quad (15)$$

The inertial velocity is then

$$v_a(x, t) = \dot{X}_a + Q_a(\Omega_a \times y + \nabla_y \phi_a). \quad (16)$$

If

$$F_a(y, t) = |y| - \mathcal{R}_a(\hat{y}, t), \quad (17)$$

the kinematic boundary condition is

$$\partial_t F_a + \nabla_y F_a \cdot \nabla_y \phi_a = 0 \quad \text{on } F_a = 0. \quad (18)$$

E. Exact energies

The orbital kinetic energy is

$$T_{\text{orb}} = \sum_a \frac{1}{2} M_a |\dot{X}_a|^2. \quad (19)$$

The internal-plus-spin kinetic energy is

$$T_{\text{int+spin}} = \sum_a \frac{\rho_a}{2} \int_{V_a(t)} |\Omega_a \times y + \nabla \phi_a|^2 d^3y. \quad (20)$$

The self-gravity is

$$U_a^{\text{self}} = -\frac{G\rho_a^2}{2} \int_{V_a(t)} \int_{V_a(t)} \frac{d^3y d^3z}{|y - z|}. \quad (21)$$

A material or restoring energy may be added as a functional of the shape:

$$U_a^{\text{mat}} = U_a^{\text{mat}}[u_a]. \quad (22)$$

The exact mutual interaction is

$$U_{ab} = -G\rho_a\rho_b \int_{V_a(t)} \int_{V_b(t)} \frac{d^3y d^3z}{|X_a - X_b + Q_a y - Q_b z|}. \quad (23)$$

The exact Lagrangian is

$$L = T_{\text{orb}} + T_{\text{int+spin}} - \sum_a (U_a^{\text{self}} + U_a^{\text{mat}}) - \sum_{a < b} U_{ab}. \quad (24)$$

This exact moving-boundary model is conceptually useful, but not practical as a general-purpose tide integrator. The moving-domain Laplace problem, the nonlinear volume constraints, and the double volume integrals make it far more expensive than desired. The rest of the paper develops a feasible approximation that keeps the same orbit–spin–shape structure.

IV. FEASIBLE FULL MODAL MODEL

A. STF mode variables

For actual computation, it is more convenient to replace the spherical harmonic amplitudes by body-frame symmetric trace-free (STF) tensors. For each degree $\ell \geq 2$, let

$$A_a^{(L)}(t) \quad (25)$$

be a body-frame STF tensor of rank ℓ . This is equivalent to the set of coefficients $\{q_{a\ell m}\}_{m=-\ell}^{\ell}$.

The corresponding mass multipole is written as

$$M_a^{(L)} = \kappa_{a\ell} A_a^{(L)}. \quad (26)$$

For later use, recall that for any rank-2 tensor B its symmetric trace-free part is

$$\text{STF}(B) = \frac{1}{2}(B + B^\top) - \frac{1}{3}(\text{tr } B)I_3. \quad (27)$$

In particular, since $\Omega_a \Omega_a^\top$ is already symmetric, the symmetrization step in (27) is redundant and one simply has

$$\text{STF}(\Omega_a \Omega_a^\top) = \Omega_a \Omega_a^\top - \frac{1}{3}|\Omega_a|^2 I_3. \quad (28)$$

B. Corotational derivative

Because the modes are stored in the rotating body frame, the natural time derivative is the corotational derivative

$$(D_t A)_{i_1 \dots i_\ell} = \dot{A}_{i_1 \dots i_\ell} + \sum_{k=1}^{\ell} \epsilon_{i_k p q} \Omega_{a,p} A_{i_1 \dots q \dots i_\ell}. \quad (29)$$

Its second derivative is

$$D_t^2 A = D_t(D_t A). \quad (30)$$

C. External tidal moments

Let $\Phi_{\text{ext},a}$ be the potential generated by all bodies other than a . The inertial STF tidal moments at X_a are

$$E_{a,\text{in}}^{(L)} = \left[\nabla^{(L)} \Phi_{\text{ext},a}(x) \right]_{x=X_a}. \quad (31)$$

The body-frame moments are

$$E_{a,\text{body}}^{(L)} = Q_a^\top \cdots Q_a^\top E_{a,\text{in}}^{(L)}. \quad (32)$$

D. Shape-dependent spin inertia

A key improvement over the simplest tide models is that the spin inertia is allowed to depend on the deformation. Denote the body-frame spin inertia by

$$J_a = J_a[A]. \quad (33)$$

For small deformations, this may be expanded linearly in the retained modes. In particular, in the quadrupole sector one will use

$$J_a(S_a) = J_{a0} I_3 - c_{J,a} S_a + O(S_a^2). \quad (34)$$

Though this form is sufficient at the present level, an effective coefficient will be introduced later in the reduced quadrupole closure to absorb the contribution from rotational flattening.

This correction is important for leading-order consistency between tidal torques and spin dynamics. In the reduced quadrupole closure developed below, the same shape dependence is also used to represent the retained rotational-flattening response, rather than introducing a second independent conservative storage term at the same order.

E. Generalized rheology

A single linear damping term corresponds essentially to a constant time-lag response. To allow more flexible frequency dependence while keeping the system in ODE form, introduce memory variables

$$Z_{a\ell n}^{(L)}(t), \quad n = 1, \dots, N_{r,a\ell}, \quad (35)$$

where n indexes relaxation branches.

Each branch has stiffness $c_{a\ell n} \geq 0$ and relaxation time $\tau_{a\ell n} > 0$.

F. Reduced full-model Lagrangian

A convenient all-mode reduced Lagrangian is

$$L = \sum_a \left[\frac{1}{2} M_a |\dot{X}_a|^2 + \frac{1}{2} \Omega_a^\top J_a[A] \Omega_a + \sum_{\ell \geq 2} \frac{\mu_{a\ell}}{2} \|D_t A_a^{(L)}\|^2 - \sum_{\ell \geq 2} \frac{\mu_{a\ell} \omega_{a\ell}^2}{2} \|A_a^{(L)}\|^2 \right] - U(X, Q, A), \quad (36)$$

where $\mu_{a\ell}$ is a modal inertia and $\omega_{a\ell}$ is the reference mode frequency.

The interaction energy is approximated by

$$U = - \sum_{a < b} \frac{G M_a M_b}{|X_a - X_b|} + U_{\text{tide}} + U_{\text{rot}}, \quad (37)$$

with

$$U_{\text{tide}} = - \sum_a \sum_{\ell \geq 2} \frac{1}{\ell!} M_{a,\text{in}}^{(L)} E_{a,\text{in}}^{(L)}, \quad (38)$$

and

$$U_{\text{rot}} = \sum_a \beta_{a2} A_a^{(ij)} \text{STF}(\Omega_{ai} \Omega_{aj}), \quad (39)$$

where only the $\ell = 2$ sector contributes. At the full modal level this term is the conservative rotational coupling whose gradient with respect to the quadrupole mode produces the centrifugal forcing retained in the shape equation.

G. All-mode evolution equations

The reduced full-model equations are

$$M_a \ddot{X}_a = -\nabla_{X_a} U, \quad (40)$$

$$\dot{Q}_a = Q_a \hat{\Omega}_a, \quad (41)$$

and

$$\dot{\Pi}_a + \Omega_a \times \Pi_a = \tau_a, \quad \Pi_a = \frac{\partial L}{\partial \Omega_a}, \quad (42)$$

where Π_a is the total body-frame angular momentum retained by the model.

For the shape modes, use

$$\begin{aligned} \mu_{a\ell} D_t^2 A_a^{(L)} + \mu_{a\ell} \omega_{a\ell}^2 A_a^{(L)} + \sum_{n=1}^{N_{r,a\ell}} c_{a\ell n} (A_a^{(L)} - Z_{a\ell n}^{(L)}) \\ = -\kappa_{a\ell} E_{a,\text{body}}^{(L)} - \delta_{\ell 2} \beta_{a2} \text{STF}(\Omega_a \Omega_a^\top)^{(L)}. \end{aligned} \quad (43)$$

The relaxation branches satisfy

$$\tau_{a\ell n} D_t Z_{a\ell n}^{(L)} = A_a^{(L)} - Z_{a\ell n}^{(L)}. \quad (44)$$

Equation (43) contains three distinct sources of deformation: the body's own restoring tendency, the external tide, and the centrifugal flattening term in the $\ell = 2$ sector. In the rank-2 case, the rotational forcing is exactly the STF tensor defined in (28), which removes the isotropic part and retains only the pure quadrupolar flattening. In the full modal model this rotational term may be viewed as arising from the conservative coupling U_{rot} . In the reduced quadrupole closure used later, the same effect is folded into the shape-dependent inertia coefficient so that the reduced spin equation and the reduced energy remain consistent.

H. Comments on angular momentum

The angular-momentum bookkeeping of the reduced model deserves emphasis, because it is the point where a naive implementation most easily goes wrong. The reduced model is most consistent when the spin equation is derived from the same Lagrangian that defines the mode kinetic energy. In that case Π_a includes not only the rigid-spin part $J_a[A]\Omega_a$, but also the angular momentum carried by the corotational mode kinetic terms. For the quadrupole reduction developed below, retaining the leading contribution of that modal angular momentum turns out to be essential: it is the minimal correction that allows the explicit quadrupole torque to act in the spin balance without spoiling the conservative bookkeeping. Higher-order modal and higher-multipole contributions to Π_a may still be neglected in a reduced implementation, but the leading quadrupole piece should be kept whenever one wants the reduced model to represent secular obliquity evolution credibly.

V. PHYSICAL INTERPRETATION OF THE FULL MODAL MODEL

The full modal model contains the main qualitative tidal effects.

A. Tidal bulges

In the slow-forcing limit, the modes approach a quasi-static response proportional to the external tidal field:

$$A_a^{(L)} \approx -\frac{\kappa_{a\ell}}{\mu_{a\ell}\omega_{a\ell}^2} E_{a,\text{body}}^{(L)}, \quad (45)$$

up to corrections from rheology and rotation. This is the tidal bulge.

B. Rotational flattening

Even without an external perturber, the reduced energy contains the spin–shape coupling

$$U_{\text{rot},a} = \beta_{a2} A_a^{(ij)} \text{STF}(\Omega_{ai}\Omega_{aj}), \quad (46)$$

whose gradient generates the term

$$-\delta_{\ell 2} \beta_{a2} \text{STF}(\Omega_a \Omega_a^\top) \quad (47)$$

in (43). Thus an isolated rapidly rotating body does not relax to a perfect sphere; instead it approaches the rotationally flattened equilibrium selected by this conservative coupling, modified by any dissipation present.

C. Lag, synchronization, and circularization

The relaxation branches cause the deformation to lag behind the forcing. The lagged bulge exerts a torque on the spin and modifies the force on the orbit. As a result, the model supports spin synchronization or pseudo-synchronization, obliquity damping, orbital circularization, and exchange of angular momentum between orbit, spin, and tides. In the quadrupole truncation developed in the next section, these effects arise from the combined action of the lagged shape tensor, the explicit quadrupole torque, and the modal angular momentum $\Pi_{S,a}$ retained in the spin balance.

VI. QUADRUPOLE REDUCTION

The most useful implementation model keeps only the $\ell = 2$ sector.

A. Quadrupole shape tensor

Represent the body-frame quadrupole deformation by a symmetric trace-free matrix

$$S_a(t) \in \mathbb{R}^{3 \times 3}, \quad S_a = S_a^\top, \quad \text{tr } S_a = 0. \quad (48)$$

The perturbed surface is

$$r = R_a(1 + \hat{n}^\top S_a \hat{n}). \quad (49)$$

B. Quadrupole moment and inertia

The inertial mass quadrupole is

$$I_a = c_{Q,a} Q_a S_a Q_a^\top. \quad (50)$$

For the normalization used in many reduced models,

$$c_{Q,a} = \frac{2}{5} M_a R_a^2. \quad (51)$$

The body-frame spin inertia is taken as

$$J_a(S_a) = J_{a0} I_3 - \tilde{c}_{J,a} S_a, \quad (52)$$

where J_{a0} is the spherical reference inertia and $\tilde{c}_{J,a}$ is the effective reduced spin–shape coefficient. If one starts from a split into a direct inertia–response coefficient $c_{J,a}$ and a separate conservative rotational-flattening coefficient α_a , then the reduced quadrupole closure requires the combination

$$\tilde{c}_{J,a} = c_{J,a} + 2\alpha_a. \quad (53)$$

For a homogeneous reference body,

$$J_{a0} = \frac{2}{5} M_a R_a^2. \quad (54)$$

Equation (52) is a linearized leading-order correction. Because S_a is trace-free, this closure keeps $\text{tr } J_a = 3J_{a0}$ fixed. That should be understood as a small-deformation linearization: $O(S_a^2)$ corrections would in general modify the trace, so (52) is not exact for large deformation. Within the reduced quadrupole closure, however, (52) is the correct place to retain the leading rotational-flattening feedback, because the same coefficient must appear both in the spin kinetic energy and in the rotational forcing of the shape equation.

C. External tidal tensor

Let

$$D_{ab} = X_b - X_a, \quad d_{ab} = |D_{ab}|, \quad \hat{D}_{ab} = D_{ab}/d_{ab}. \quad (55)$$

The inertial quadrupolar tidal tensor produced by the other bodies is

$$E_{a,\text{in}} = \sum_{b \neq a} \frac{GM_b}{d_{ab}^3} \left(I_3 - 3\hat{D}_{ab}\hat{D}_{ab}^\top \right). \quad (56)$$

The body-frame tidal tensor is

$$E_{a,\text{body}} = Q_a^\top E_{a,\text{in}} Q_a. \quad (57)$$

Because the quadrupole interaction energy is

$$U_{\text{tide},a}^{(2)} = +\frac{1}{2} I_a : E_{a,\text{in}} = +\frac{c_{Q,a}}{2} S_a : E_{a,\text{body}}, \quad (58)$$

the reciprocal tidal forcing that appears in the shape equation is

$$\lambda_a E_{a,\text{body}}, \quad \lambda_a \equiv \frac{c_{Q,a}}{2}. \quad (59)$$

This coefficient is required for consistency between the quadrupole force, torque, interaction energy, and the internal mode dynamics.

In the reduced quadrupole closure adopted here, rotational flattening is not kept as a separate stored term $U_{\text{rot},a}^{(2)}$. Instead it is absorbed into the same linearized inertia coefficient that appears in (52). Because S_a is symmetric and trace-free,

$$\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a = \frac{1}{2} J_{a0} |\Omega_a|^2 - \frac{1}{2} \tilde{c}_{J,a} S_a : \text{STF}(\Omega_a \Omega_a^\top), \quad (60)$$

so the rotational contribution implied by the reduced spin kinetic energy is

$$-\frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top). \quad (61)$$

This is the coefficient that must appear in the reduced quadrupole shape equation. Adding a second independent term $\alpha_a S_a : \text{STF}(\Omega_a \Omega_a^\top)$ to the reduced mechanical energy would double count the same conservative coupling unless one also kept the corresponding extra contribution to the reduced spin momentum.

D. Corotational derivative

For any body-frame rank-2 tensor, define

$$D_t S_a = \dot{S}_a + [\hat{\Omega}_a, S_a], \quad (62)$$

where

$$[A, B] = AB - BA. \quad (63)$$

Then

$$D_t^2 S_a = D_t(D_t S_a). \quad (64)$$

It is convenient to introduce the auxiliary variable

$$W_a = D_t S_a. \quad (65)$$

E. Quadrupole shape equation with rheology

The quadrupole shape equation is

$$D_t^2 S_a + 2\gamma_a D_t S_a + \omega_{a2}^2 S_a + \sum_{n=1}^{N_{r,a}} c_{an} (S_a - Z_{an}) = -\lambda_a E_{a,\text{body}} - \frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top). \quad (66)$$

Each relaxation branch obeys

$$\tau_{an} D_t Z_{an} = S_a - Z_{an}. \quad (67)$$

The coefficient γ_a represents direct viscous-like damping of the quadrupole rate, while the branch terms represent delayed internal response. A single branch is often enough for a useful toy model; several branches can approximate a broader frequency-dependent rheology.

F. Spin equation

For obliquity-sensitive applications, the reduced quadrupole model should retain the leading angular momentum carried by the quadrupole mode itself. With

$$W_a = D_t S_a, \quad (68)$$

and with $\text{axl}(\hat{v}) = v$ for antisymmetric matrices, the leading quadrupole modal contribution is

$$\Pi_{S,a} = 2 \text{axl}([W_a, S_a]). \quad (69)$$

This follows directly from $\Pi_a = \partial L / \partial \Omega_a$ applied to the quadrupole kinetic term $\frac{1}{2} \|W_a\|^2$.

The same objective-rate dissipative terms that appear in the shape equations also generate reciprocal body torques. For the direct viscous damping and the relaxation branches, define

$$N_{W,a}^{\text{diss}} = -2\gamma_a \Pi_{S,a} = -4\gamma_a \text{axl}([W_a, S_a]), \quad (70)$$

and

$$N_{Z,a}^{\text{diss}} = -2 \sum_{n=1}^{N_{r,a}} c_{an} \text{axl}([S_a - Z_{an}, Z_{an}]). \quad (71)$$

These terms vanish in the nondissipative limit and ensure that the same objective-rate dissipation that damps the internal tensors also removes the corresponding body-frame angular momentum from the rigid-spin sector.

The improved body-frame quadrupole spin equation is therefore

$$\frac{d}{dt} \left(J_a(S_a) \Omega_a + \Pi_{S,a} \right) + \Omega_a \times \left(J_a(S_a) \Omega_a + \Pi_{S,a} \right) = N_{a,\text{body}} + N_{W,a}^{\text{diss}} + N_{Z,a}^{\text{diss}}. \quad (72)$$

Expanding the time derivative gives

$$J_a(S_a) \dot{\Omega}_a = N_{a,\text{body}} + N_{W,a}^{\text{diss}} + N_{Z,a}^{\text{diss}} - \dot{J}_a(S_a) \Omega_a - \dot{\Pi}_{S,a} - \Omega_a \times \left(J_a(S_a) \Omega_a + \Pi_{S,a} \right). \quad (73)$$

Since $W_a = D_t S_a$, one also has

$$\dot{S}_a = W_a - [\hat{\Omega}_a, S_a], \quad (74)$$

so that in the linearized closure

$$\dot{J}_a(S_a) = -\tilde{c}_{J,a} \dot{S}_a. \quad (75)$$

The modal contribution evolves as

$$\dot{\Pi}_{S,a} = 2 \text{axl}([\dot{W}_a, S_a] + [W_a, \dot{S}_a]). \quad (76)$$

Equations (72)–(76) explicitly retain the orientational dynamics carried by the quadrupole mode. The shape dependence of $J_a(S_a)$ still captures the accompanying change in spin inertia, but retaining $\Pi_{S,a}$ is required to close the spin–tide angular-momentum exchange at this order.

G. Quadrupole torque

The inertial quadrupole torque associated with the same truncated interaction energy is

$$N_{a,\text{in}} = 3 \sum_{b \neq a} \frac{GM_b}{d_{ab}^3} (I_a \hat{D}_{ab}) \times \hat{D}_{ab}. \quad (77)$$

The body-frame version is

$$N_{a,\text{body}} = Q_a^\top N_{a,\text{in}}. \quad (78)$$

In the improved quadrupole closure, (77)–(78) are not merely diagnostic: they are the physical right-hand side of (72). Because the spin balance explicitly retains the modal angular momentum (69) generated by the same corotational kinematics, this torque enters without double-counting the conservative Q_a – S_a coupling.

H. Orbital dynamics

The pair interaction energy truncated at monopole plus quadrupole order is

$$U_{ab} = -\frac{GM_a M_b}{d_{ab}} - \frac{3G}{2d_{ab}^3} (M_b \hat{D}_{ab}^\top I_a \hat{D}_{ab} + M_a \hat{D}_{ab}^\top I_b \hat{D}_{ab}). \quad (79)$$

This retains the monopole and monopole–quadrupole couplings, but neglects quadrupole–quadrupole interactions of order $O(d_{ab}^{-5})$ and all higher multipolar contributions.

The orbital equation is

$$M_a \ddot{X}_a = -\nabla_{X_a} \sum_{b < c} U_{bc}. \quad (80)$$

Equations (1), (66), (67), (72), and (80) form a closed quadrupole tide model. Relative to the simpler reduced closure that keeps only $J_a(S_a)\Omega_a$, the additional term (69) is algebraically modest but physically important for obliquity evolution.

I. First-order implementation form

Using the auxiliary variable $W_a = D_t S_a$, the quadrupole subsystem may be written in first-order form as

$$D_t S_a = W_a, \quad (81)$$

$$D_t W_a = -2\gamma_a W_a - \omega_{a2}^2 S_a - \sum_{n=1}^{N_{r,a}} c_{an}(S_a - Z_{an}) - \lambda_a E_{a,\text{body}} - \frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top), \quad (82)$$

together with (67). This form is convenient for standard ODE solvers. In particular, if one implements the inertial quadrupole as $I_a = c_{Q,a} Q_a S_a Q_a^\top$ in the orbital force and pair potential, then the same model must use the reciprocal coefficient $\lambda_a = c_{Q,a}/2$ in (66) and (82); otherwise the bookkeeping of conservative energy is inconsistent. Likewise, the same coefficient $\tilde{c}_{J,a}$ must appear both in the inertia tensor (52) and in the rotational forcing term of (66) and (82). In the improved closure, the spin update should then be performed from (73) with the explicit conservative torque $N_{a,\text{body}}$, the reciprocal dissipative torques (70)–(71), and the modal angular-momentum term (69). This is still a modest implementation change because $\dot{S}_a$, $\dot{W}_a$, and the branch lags $S_a - Z_{an}$ are already available from the first-order state, so both $\dot{\Pi}_{S,a}$ in (76) and the algebraic dissipative torques can be assembled at the same right-hand-side evaluation. In the split extension below, the same update simply includes the additional bulge-branch torque (93). No additional conservative storage term proportional to $S_a : \text{STF}(\Omega_a \Omega_a^\top)$ should be added on top of $\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a$, because that Hamiltonian term already contains the rotational-flattening coupling in this reduced closure.

J. Conservative quadrupole energy

In the nondissipative limit $\gamma_a = 0$ and with all relaxation branches removed, the natural mechanical energy of the quadrupole closure is

$$E_{\text{mech}} = \sum_a \left[\frac{1}{2} M_a |\dot{X}_a|^2 + \frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a + \frac{1}{2} \|W_a\|^2 + \frac{1}{2} \omega_{a2}^2 \|S_a\|^2 \right] + \sum_{a < b} U_{ab}, \quad (83)$$

up to the same truncation order used elsewhere in the reduced model and with any chosen normalization of the STF norm. The rotational-flattening coupling is already contained in $\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a$ through the shape dependence of $J_a(S_a)$. Therefore the reduced conservative diagnostic must not add a second term proportional to $S_a : \text{STF}(\Omega_a \Omega_a^\top)$ unless the reduced spin equation is simultaneously upgraded to evolve the corresponding extra contribution to the body-frame angular momentum.

VII. SPLIT ROTATIONAL-BULGE EXTENSION FOR OBLIQUITY DAMPING

The one-tensor quadrupole closure developed above formally conserves angular momentum by retaining the modal contribution $\Pi_{S,a}$ and the explicit quadrupole torque in the spin balance. Even so, a single STF tensor S_a must then represent both the tide-raised bulge and the rotationally supported oblateness. These two mechanisms generally relax on different timescales and need not remain aligned with the same direction, so a more flexible reduced model separates them.

A simple next-level reduced model is therefore to split the quadrupole into a *tide-driven* STF tensor S_a and a separate

rotational bulge B_a , also taken symmetric and trace free. The total body-frame quadrupole is then written as

$$I_{a,\text{body}} = c_{Q,a}^{(t)} S_a + c_{Q,a}^{(r)} B_a, \quad (84)$$

and the corresponding inertial quadrupole is therefore

$$I_a = Q_a I_{a,\text{body}} Q_a^\top = Q_a (c_{Q,a}^{(t)} S_a + c_{Q,a}^{(r)} B_a) Q_a^\top, \quad (85)$$

while the body-frame spin inertia becomes

$$J_a(B_a) = J_{a0} I_3 - c_{J,a}^{(r)} B_a. \quad (86)$$

In this split closure the tide-driven tensor S_a contributes to the external quadrupole but not directly to the reduced spin inertia. This avoids reintroducing a separate conservative spin–shape coupling into the S_a equation while keeping the rotational bulge B_a as the explicit carrier of spin-supported flattening. The tide-driven quadrupole continues to satisfy the second-order rheological shape equation,

$$\begin{aligned} D_t^2 S_a + 2\gamma_a D_t S_a + \omega_{a2}^2 S_a + \sum_{n=1}^{N_{f,a}} c_{an} (S_a - Z_{an}) \\ = -\lambda_a E_{a,\text{body}}, \end{aligned} \quad (87)$$

with the same branch equations

$$\tau_{an} D_t Z_{an} = S_a - Z_{an}. \quad (88)$$

The rotational bulge is promoted to its own reduced variable and evolved with a first-order corotating relaxation law,

$$\tau_{B,a} D_t B_a + B_a = -\eta_a \text{STF}(\Omega_a \Omega_a^\top) - \lambda_{B,a} E_{a,\text{body}}, \quad (89a)$$

$$\lambda_{B,a} = \eta_a \frac{c_{J,a}^{(r)}}{c_{J,a}^{(r)}}, \quad (89b)$$

where $\tau_{B,a}$ is the relaxation time of the spin-supported figure and $\eta_a > 0$ controls its quasi-static amplitude. The additional reciprocal tidal term is required if the same B_a contributes both to the spin inertia and to the external quadrupole: otherwise the orbital field can do conservative work on B_a through I_a with no matching forcing in the B_a equation. With the STF sign convention used throughout this paper, the minus sign in the spin forcing is required for a positive-spin-rate hydrostatic bulge to be oblate, i.e. compressed along the spin axis and enlarged in the orthogonal directions. In the limit $\tau_{B,a} \rightarrow 0$ the spin bulge approaches the instantaneous hydrostatic figure, while finite $\tau_{B,a}$ allows the rotational bulge to lag behind changes in spin rate, spin direction, or the body-frame tidal tensor. Because B_a is itself a shape tensor, the coefficient $c_{J,a}^{(r)}$ in (86) should be read as the geometric map from shape to spin inertia. It is typically of the same order as the ordinary inertia-shape coefficient $c_{J,a}$. The extra rotational-flattening strength that was merged into $\tilde{c}_{J,a} = c_{J,a} + 2\alpha_a$ in the one-tensor closure should here be absorbed into the response amplitude η_a , not into an additional direct S_a -dependence of the spin inertia.

For diagnostics and energy bookkeeping in the split closure, the stored mechanical contribution of the first-order bulge branch is the non-negative self-energy

$$\mathcal{E}_B = \sum_a \frac{c_{J,a}^{(r)}}{4\eta_a} B_a : B_a. \quad (90)$$

No additional explicit $B_a : E_{a,\text{body}}$ term should be added on top of this: the conservative tidal coupling of B_a is already present through the total quadrupole in the pair interaction energy (79), and adding it again would double count the same interaction.

Matching the low-frequency visible equilibrium of the one-tensor model gives $\eta_a \approx (c_{J,a} + 2\alpha_a)/(2\omega_{a2}^2)$, while matching its low-frequency phase lag gives $\tau_{B,a} \approx 2\gamma_a/\omega_{a2}^2$.

For bookkeeping, the non-negative dissipation rate of this first-order branch is naturally governed by the lag away from the instantaneous target $B_{a,*} = -\eta_a \text{STF}(\Omega_a \Omega_a^\top) - \lambda_{B,a} E_{a,\text{body}}$, namely

$$\dot{D}_B = \sum_a \frac{c_{J,a}^{(r)}}{2\eta_a \tau_{B,a}} (B_a - B_{a,*}) : (B_a - B_{a,*}). \quad (91)$$

The recoverable mechanical energy is therefore the branch *self-energy* (90), not the lag energy. Equivalently, completing the square around $B_{a,*}$ produces a lag-squared term plus a forcing-dependent offset. Keeping only $(B_a - B_{a,*}) : (B_a - B_{a,*})$ as a stored-energy diagnostic would make the bookkeeping look artificially non-conservative in oblique or precessing runs because $B_{a,*}$ changes as the body-frame tidal tensor and spin direction evolve.

The spin equation is then upgraded only slightly:

$$\begin{aligned} \frac{d}{dt} \left(J_a(B_a) \Omega_a + \Pi_{S,a} \right) + \Omega_a \times \left(J_a(B_a) \Omega_a + \Pi_{S,a} \right) = \\ N_{a,\text{body}} + N_{W,a}^{\text{diss}} + N_{Z,a}^{\text{diss}} + N_{B,a}^{\text{diss}}, \end{aligned} \quad (92)$$

where $\Pi_{S,a} = 2 \text{axl}([W_a, S_a])$ is still the explicit modal angular momentum retained from the second-order tide-driven quadrupole, and the first-order bulge branch contributes the reciprocal dissipative torque

$$N_{B,a}^{\text{diss}} = -\frac{c_{J,a}^{(r)}}{\eta_a} \text{axl}([B_{a,*} - B_a, B_a]), \quad (93a)$$

$$B_{a,*} = -\eta_a \text{STF}(\Omega_a \Omega_a^\top) - \lambda_{B,a} E_{a,\text{body}}. \quad (93b)$$

The first-order variable B_a does not introduce an additional second-order modal kinetic term, so no separate $\Pi_{B,a}$ is required in this reduced extension. The orbital force and body-frame torque are computed from the total quadrupole (85) in the same way as before. In particular, the tide-driven shape equation continues to use only the reciprocal coupling to the tide-driven piece, namely $-\lambda_a E_{a,\text{body}}$ with $\lambda_a = c_{Q,a}^{(t)}/2$; the rotational bulge enters the orbital dynamics and torque only through the total quadrupole I_a , the spin inertia $J_a(B_a)$, and the branch torque $N_{B,a}^{\text{diss}}$ associated with its corotational relaxation.

This split extension remains simple compared with the rest of the model. It adds only one extra STF tensor and one extra relaxation timescale per body, and it does not alter the pairwise $O(N^2)$ force evaluation. The expected benefit is specifically in spin-axis evolution: the torque acting on the separately relaxing rotational bulge can dissipate precessional energy even when the tide-driven quadrupole is dominated by the line-of-centers forcing. That makes robust obliquity damping a more realistic outcome of the reduced model.

VIII. RELATION TO REAL PROBLEMS

A. What remains toy-like

The model retains the key qualitative features of tidal evolution—full three-dimensional orbital motion, explicit tidal multipoles and their backreaction, spin–orbit coupling, rotational flattening, and frequency-dependent dissipative lag—while remaining in ODE form. This suffices to reproduce the leading-order structure of spin synchronization, orbital circularization, and obliquity damping qualitatively. However, several important simplifications remain.

The interior is not evolved hydrodynamically, and real planets may contain cores, mantles, oceans, and layered viscosities that are not represented here. The rheological coefficients γ_a , c_{an} , and τ_{an} are effective reduced parameters representing a linearized frequency response, not a detailed microphysical material law. The model retains only the leading quadrupole angular-momentum correction $\Pi_{S,a}$; higher-order modal contributions and a fully systematic continuum angular-momentum closure are still omitted, so exact global angular-momentum conservation holds only within the retained truncation order. The linearized inertia (52) is intended for weak to moderate deformations; very strong tides would require higher modes and nonlinear corrections. Mechanical dissipation is included, but heating, melting, convection, and feedback on material parameters are not.

B. Comparison with common tide models

A single damping term proportional to $D_t S_a$ resembles a constant time-lag description. By contrast, the branch variables Z_{an} allow a frequency-dependent lag more like a generalized Maxwell or anelastic response. This is still a reduced rheology, but it is more flexible and more realistic than a single constant-lag parameter.

C. Relation to Love number and quality factor

Planetary tidal dissipation is often parameterized by the quadrupolar Love number k_2 and the quality factor Q . In the present reduced model, the more fundamental objects are the linear response coefficients appearing in (66). For a harmonic forcing at angular frequency σ in the body frame, one may write the complex quadrupole response schematically as

$$\tilde{S}_a(\sigma) = -\chi_a(\sigma)\tilde{F}_a(\sigma), \quad (94)$$

where $\tilde{F}_a$ denotes the STF forcing from the tidal tensor and the rotational term, and where

$$\chi_a(\sigma) = \left[-\sigma^2 + 2i\gamma_a\sigma + \omega_{a2}^2 + \sum_{n=1}^{N_{r,a}} c_{an} \left(1 - \frac{1}{1 + i\sigma\tau_{an}} \right) \right]^{-1} \quad (95)$$

in the linearized frequency-domain approximation. The static compliance $\chi_a(0)$ sets the scale of the equilibrium quadrupole and therefore of an effective static k_2 , while the ratio of imaginary to real parts of $\chi_a(\sigma)$ determines the phase lag and hence an effective $Q^{-1}(\sigma)$.

The present parameters $(\gamma_a, \omega_{a2}, c_{an}, \tau_{an})$ should therefore be viewed as a way to fit or calibrate the complex quadrupolar

response of a body over the frequency range of interest, rather than as a unique microscopic material model. With one branch the model resembles a simple constant-lag-like response over a limited band, whereas several branches can be chosen to better match a desired frequency-dependent $k_2(\sigma)$ and $Q(\sigma)$.

IX. IMPLEMENTATION AND COMPUTATIONAL REMARKS

The upgraded quadrupole model remains inexpensive compared with full hydrodynamics. For N bodies, the dominant cost of each right-hand-side evaluation is the pairwise gravitational and tidal interaction, scaling as $O(N^2)$. The additional costs from the refinements are all lower order: building and inverting the 3×3 inertia tensor $J_a(S_a)$ is $O(N)$, the rotational-flattening term is $O(N)$, and $N_{r,a}$ rheology branches add $O(NN_r)$ state variables and arithmetic. The main effect of the improvements is therefore a modest increase in state size and per-body algebra, not a qualitative change in complexity.

The accompanying reference code integrates the quadrupole closure directly. For body a the evolved state is

$$(X_a, \dot{X}_a, q_a, \Omega_a, S_a, W_a, Z_{a1}, \dots, Z_{aN_{r,a}}), \quad (96)$$

with an additional split-bulge tensor B_a when the split rotational-bulge extension is enabled. Here q_a is a unit quaternion representing the body orientation.

The key numerical choices are the following:

- orientations are advanced with quaternions and renormalized as needed,
- S_a, W_a, Z_{an} , and when present B_a are stored as full 3×3 matrices and reprojected onto their symmetric trace-free parts to suppress numerical drift,
- the orbital force, pair potential, and quadrupole torque are assembled from the same truncated potential, while the spin equation also retains the quadrupole-mode angular momentum term $\Pi_{S,a} = 2 \text{axl}([W_a, S_a])$ together with the reciprocal dissipative torques generated by the active relaxation variables,
- the smallest eigenvalue of the shape-dependent inertia tensor is watched throughout the run so that the evolution stays in the weak-deformation regime; in the one-tensor closure, $J_a(S_a) = J_{a0}I_3 - \tilde{c}_{J,a}S_a$, so one must keep $\tilde{c}_{J,a}\lambda_{\max}(S_a) < J_{a0}$.

A straightforward numerical realization stores each STF quadrupole variable S_a, W_a , and Z_{an} as a full 3×3 matrix rather than in a minimal five-component basis. This is slightly redundant, but it keeps the ODEs close to their analytical form: the corotational terms remain simple commutators, and the rotational forcing is just $\text{STF}(\Omega_a \Omega_a^\top)$. Numerical drift can then be controlled by re-projecting these matrices onto their symmetric trace-free parts after right-hand-side evaluations or at output times.

The orientation may be represented by unit quaternions q_a mapping body-frame vectors into inertial-frame vectors. For moderate N , the orbital force, body-frame tidal tensor, and diagnostic quadrupole torque can all be assembled by vectorized pairwise operations, preserving the $O(N^2)$ cost of the reduced model. For the particular leading-order closure

used here, it is also convenient in the spin update to solve the 3×3 linear system

$$J_a(S_a)\dot{\Omega}_a = -\dot{J}_a(S_a)\Omega_a - \Omega_a \times (J_a(S_a)\Omega_a) \quad (97)$$

directly, rather than forming $J_a(S_a)^{-1}$ explicitly.

For generalized rheology, initializing each branch near $Z_{an} = S_a$ avoids an artificial start-up transient. The reference implementation follows the improved quadrupole closure adopted in this paper: it keeps the deformation dependence of $J_a(S_a)$, retains the leading quadrupole angular momentum $\Pi_{S,a} = 2 \text{axl}([W_a, S_a])$, and includes the torque (78) explicitly in the spin balance. The conservative diagnostic should still be constructed from the same truncated reduced energy used by the ODEs, with the same coefficient $\tilde{c}_{J,a}$ appearing in both $J_a(S_a)$ and the rotational forcing term. No additional $S_a : \text{STF}(\Omega_a \Omega_a^\top)$ storage term should be added on top of $\frac{1}{2}\Omega_a^\top J_a(S_a)\Omega_a$ in this closure. After that correction, remaining small drift in practical integrations is primarily numerical and reflects the use of finite tolerances, post-processed dissipation integrals, and the residual truncation error of the reduced model rather than an omitted conservative energy reservoir. In production runs it is also worth monitoring the smallest eigenvalue of $J_a(S_a)$ and rejecting parameter choices or time steps that drive the inertia tensor close to singularity.

X. ILLUSTRATIVE DOUBLE-PLANET EVOLUTION

To illustrate the qualitative behavior of the quadrupole model, this section presents two representative integrations for close double-planet systems. Both use the same orbital and spin configuration but differ in the material parameters of the planets. Together they exhibit qualitatively distinct regimes of tidal evolution: smooth secular decay driven by a quasi-static bulge, and impulsive dissipation concentrated in a pair of resonance-passage events.

A. Smooth dissipative evolution

Figure 1 shows an integration starting from a randomly initialized eccentric, oblique, asynchronously rotating configuration with non-zero initial quadrupole deformation. The figure is not intended as a parameter survey, but as a compact diagnostic of how the reduced model moves energy and angular momentum between orbit, spin, and tidal shape variables during long-term evolution.

The upper-left panel shows the energy bookkeeping. The mechanical energy (green dashed) and the cumulative dissipated energy (black) each vary by roughly 15% of the initial orbital energy over the course of the run, yet their compensated sum (red) remains essentially flat throughout—its total variation is well under one percent of the mechanical energy scale. That near-constancy confirms that the reduced ODE closure is accounting for dissipated energy consistently rather than leaking it into truncated degrees of freedom.

The upper-right panel reveals a striking asymmetry between the two bodies. Planet 2's squared deformation rate $\|W\|_F^2$ and lag $\|S - Z\|_F^2$ both spike to roughly 0.10 at the start, then collapse by two to three orders of magnitude within the first ~ 100 time units. Planet 1's analogous quantities are so small

they barely register on the same scale. The burst for planet 2 is essentially a rapid re-equilibration from an initial state far from tidal equilibrium. After that transient, both bodies settle into the slowly-evolving quasi-static regime in which the quadrupole closely tracks its instantaneous equilibrium value.

The lower-left panel shows the secular orbital evolution. Both the eccentricity and the semi-major axis decrease monotonically, and their curves run roughly parallel throughout, reflecting the coupled circularization–inspiral expected from tidal dissipation. The decay is not uniform: there is a noticeable acceleration around $t \approx 4000$ where both curves steepen before settling into a final decline.

The lower-right panel compares the spin angular frequencies of the two planets with the orbital mean motion. Both spin rates are nearly constant on the scale of the plot; any spin-down is small compared with the rise in orbital mean motion as the orbit contracts. The approach toward synchronization is therefore driven primarily by orbital inspiral rather than by spin-down. Planet 2, spinning at ~ 0.84 , ends the run at roughly twice the final orbital frequency—substantially closer to synchronization than planet 1, which spins at ~ 1.19 , or nearly three times the final orbital mean motion. The spin-relaxation timescale evidently exceeds the integration window by a considerable margin.

B. Resonance-passage evolution

Figure 2 shows a second integration run with the same orbital and spin initial conditions but with both planets initialized at zero deformation ($S_0 = W_0 = 0$) and with higher elastic mode stiffnesses. The resulting dynamics are qualitatively different and considerably more structured.

Because both planets start from zero deformation, there is no initial equilibration transient: the quadrupole activity begins near zero and stays there until the first event. The evolution then proceeds in three distinct phases separated by the two bursts.

Between events, quasi-static evolution. During the long quiet intervals— $t \lesssim 900$ and $1200 \lesssim t \lesssim 4300$ —the reduced dynamics are indistinguishable from ordinary quasi-static tidal friction. The bulge tracks the instantaneous tidal forcing closely ($\|W\|_F^2$ and $\|S - Z\|_F^2$ both small), the orbit circularizes and contracts gradually, and the spin rates are essentially constant. This is the regime captured by the smooth run above, but here the interlude is interrupted.

Resonance passages. As the orbit slowly contracts, the orbital mean motion n increases. Because the orbital forcing of a tidally deformable body contains harmonics of the mean motion, the effective forcing frequency near the body's natural quadrupole frequency ω_{a2} sweeps upward in time. When a harmonic kn passes through ω_{a2} , the quadrupole mode is briefly driven near resonance: the deformation amplitude rises, dissipation intensifies, and angular momentum is exchanged between the orbit and the internal degrees of freedom before the mode is detuned by the resulting orbital change. The event is self-limiting—the orbital shift moves the system off resonance—leaving a permanent step-down in eccentricity and semi-major axis. Resonance sweeping of this kind is

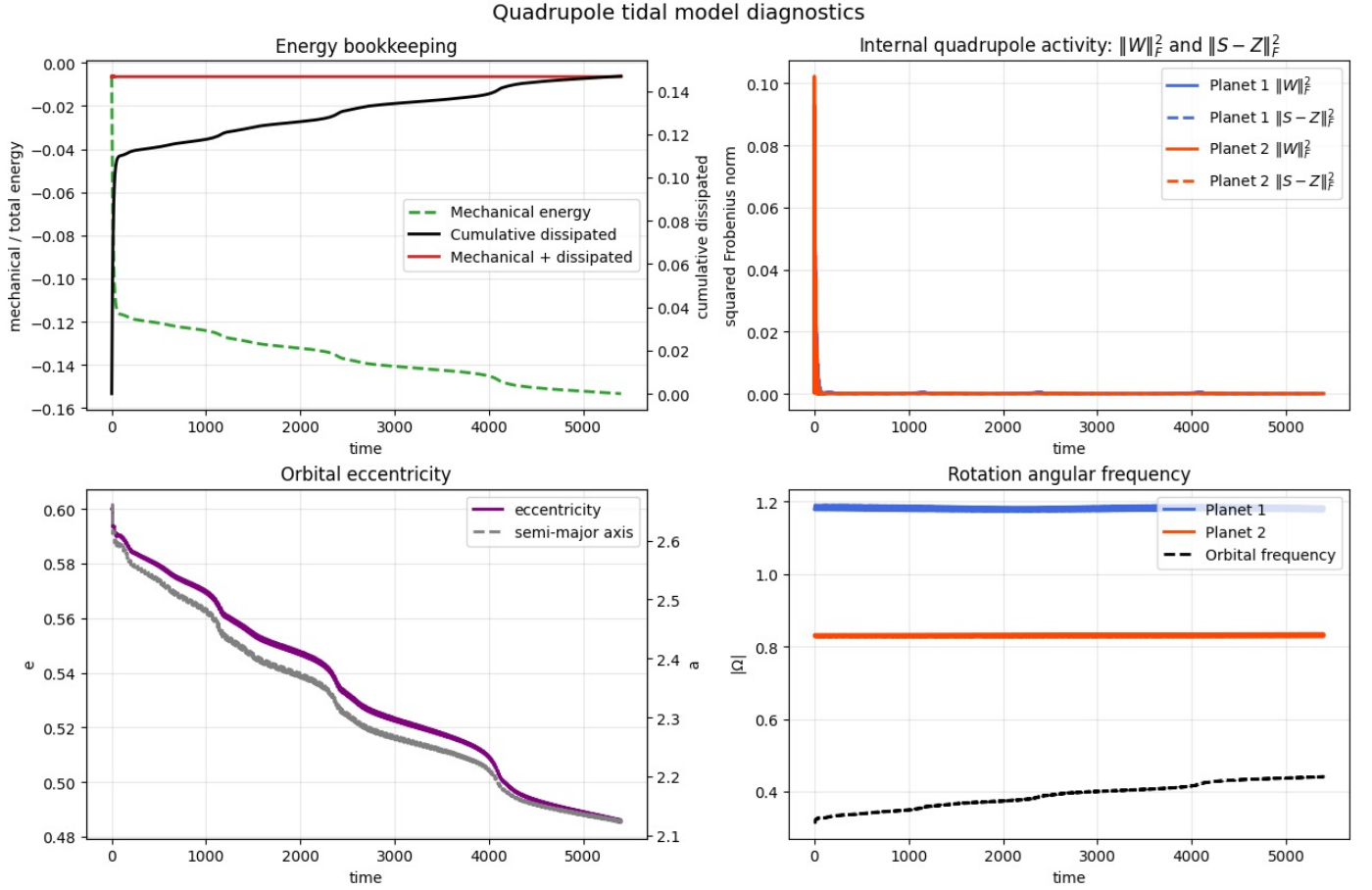


Fig. 1. Smooth dissipative double-planet integration with the split quadrupole model. Both planets start from a non-zero deformation and relax rapidly into the quasi-static tidal regime, after which energy is steadily drained from the orbit by the lagged quadrupole. The compensated mechanical-plus-dissipated energy (red) remains flat to within well under one percent of the mechanical energy scale throughout.

a recognized mechanism in celestial mechanics, though its occurrence in a reduced ODE model of this type is instructive in its own right.

In the present run, planet 1 ($\omega_{12} = 4.54$) undergoes its passage first, at $t \approx 1000$. The burst in $\|W\|_F^2$ for planet 1 (blue, upper-right panel) is large and multi-peaked, indicating that the system oscillates through the resonance over several orbital periods before settling. Planet 2 ($\omega_{22} = 5.06$) resonates later, at $t \approx 4500$, and the corresponding orange burst is similarly structured. The two mode frequencies are close but distinct, so the passages occur at different times as the mean motion climbs through the relevant harmonic.

Each event has a clear signature in all four panels simultaneously: a burst in the quadrupole activity (upper-right), a step in cumulative dissipation and a corresponding drop in mechanical energy (upper-left), and step-down features in both eccentricity and semi-major axis (lower-left). The fact that all four panels are consistent is a non-trivial check: the impulsive events are not numerical artifacts but are physically coherent within the reduced ODE closure.

Spin evolution. The lower-right panel shows that both spin rates are essentially frozen throughout, including during the bursts. The spin-down from 3.73 to 3.44 (planet 1) and 2.63 to 2.42 (planet 2) measured at the end of the run represents a

change of roughly 8% in both cases over the full integration, but this change is spread so uniformly that it is imperceptible on the scale of the panel. The orbital mean motion, by contrast, rises visibly. The resonance passages do not cause detectable kicks to the spin rate, consistent with the expectation that the torque impulse during a short resonance event is much smaller than the angular momentum stored in the spin.

Energy bookkeeping. The upper-left panel confirms that the conserved mechanical-plus-dissipated sum (red) is flat throughout—including during both burst events—to within the numerical precision of the run. The total dissipated energy reaches 8.9×10^{-3} , split roughly evenly between the two events and the quiet intervals. This is a more stringent test of the reduced closure than the smooth run, because the impulsive events require the energy bookkeeping to remain consistent during rapid transient dynamics rather than only in the quasi-static limit.

Taken together, the two integrations show that the reduced quadrupole ODE closure is capable of representing qualitatively different regimes of tidal evolution within the same framework, and that the energy bookkeeping remains mechanically consistent in both the smooth and impulsive cases.

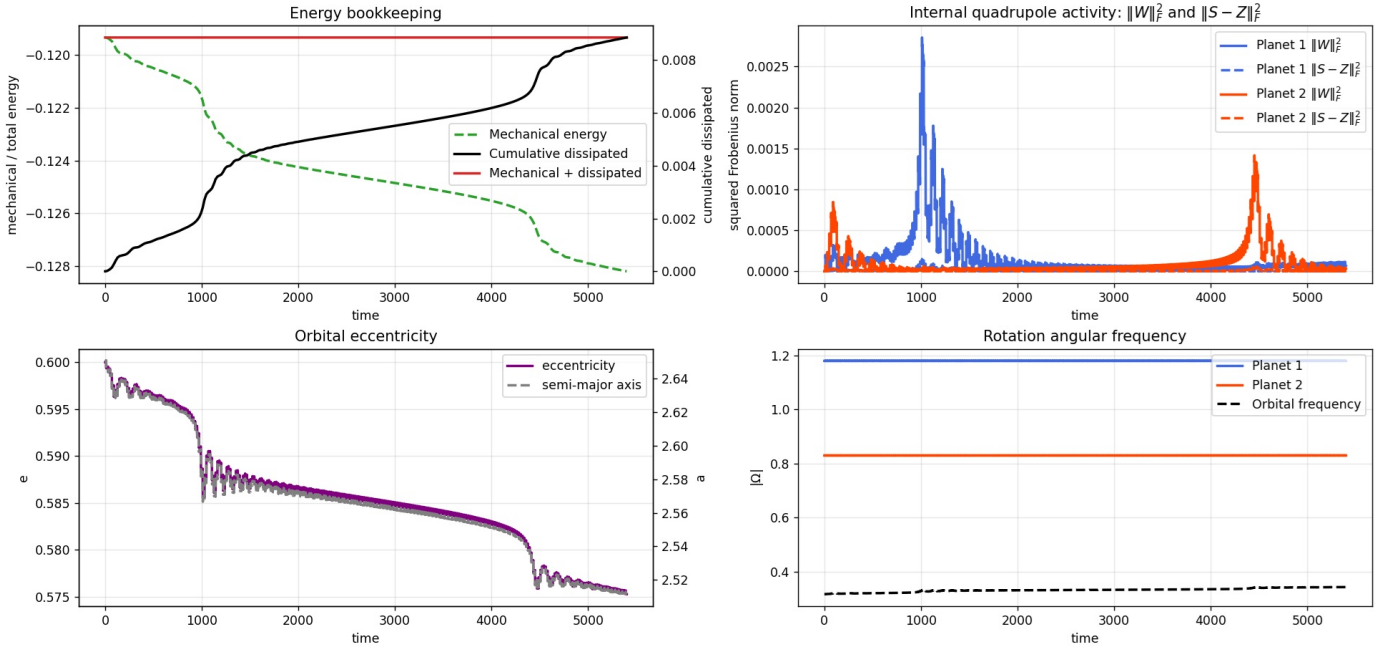


Fig. 2. Resonance-passage double-planet integration. Both planets are initialized with zero quadrupole deformation. The run reveals two isolated bursts of internal quadrupole activity—at $t \approx 1000$ (planet 1 dominant, blue) and $t \approx 4500$ (planet 2 dominant, orange)—each accompanied by a sudden step-down in eccentricity and a corresponding jump in cumulative dissipation. Between these events the system evolves quasi-statically and the spin rates remain essentially frozen. The compensated energy (red, upper-left) is flat throughout, confirming that the impulsive events are physically captured by the reduced ODE closure rather than being numerical artifacts.

XI. MODEL VERIFICATION

The reference implementation includes a regression-style driver `run_tests()` that integrates representative two-body problems, reconstructs derived orbital and spin diagnostics from the saved trajectory, and then applies explicit quantitative pass/fail thresholds. Each case also writes a diagnostic plot. The purpose of the suite is not to calibrate the model against a particular planetary system, but to verify that the reduced ODE closure behaves in the limits it was designed to capture: the null Kepler limit, conservative equilibrium, quasi-static response, rotational flattening, rheological distinguishability, secular circularization, spin synchronization, and obliquity damping.

A. Implemented checks

a) 01_null_kepler_limit: This case verifies that the code reduces to ordinary two-body Kepler motion when all quadrupole couplings are turned off.

Pass criteria. Maximum drift in eccentricity $< 5 \times 10^{-6}$, maximum drift in semi-major axis $< 5 \times 10^{-6}$, and maximum drift in the conservative energy diagnostic $< 5 \times 10^{-7}$.

Current result. All three margins are passed comfortably, with $\max |\Delta e| = 3.674 \times 10^{-9}$, $\max |\Delta a| = 1.790 \times 10^{-8}$, and $\max |\Delta E_{\text{total}}| = 7.308 \times 10^{-10}$.

b) 02_conservative_equilibrium: This case checks that a circular synchronized configuration initialized near the reduced static quadrupole equilibrium remains there in the nondissipative limit.

Pass criteria. Maximum total-energy drift $< 2 \times 10^{-5}$, maximum eccentricity $< 2 \times 10^{-3}$, and mean bulge–line-of-centers angle $< 2^\circ$.

Current result. The run passes by a wide margin with $\max |\Delta E_{\text{total}}| = 1.019 \times 10^{-14}$, $e_{\text{max}} = 2.862 \times 10^{-7}$, and mean bulge angle 0.579° .

c) 03_quasi_static_bulge: This case verifies that strongly damped quadrupoles track the instantaneous hydrostatic tide in the slow-forcing regime.

Pass criteria. Mean relative shape error in the late-time part of the run < 0.20 , final relative shape error < 0.10 , and mean bulge lag $< 5^\circ$.

Current result. The implementation passes with mean and final relative shape errors both 8.449×10^{-3} and a mean bulge lag of 0.393° .

d) 04_rotational_flattening: This case isolates the spin-supported oblateness mechanism and confirms that the shape equation reproduces the expected oblate equilibrium when the tidal field is negligible.

Pass criteria. Relative error to the isolated rotational equilibrium < 0.10 and the presence of an oblate STF eigenvalue pattern.

Current result. The run passes with relative error 3.321×10^{-9} and eigenvalues $[-0.0693, 0.0347, 0.0347]$, as expected for an oblate figure.

e) 05_rheology_comparison: This comparison confirms that the direct viscous closure and a delayed Maxwell-like branch closure are observationally distinguishable in the same orbital setup. The code implements this

as two companion runs, 05A_rheology_viscous and 05B_rheology_maxwell.

Pass criteria. The absolute difference in mean bulge lag must exceed 0.5° , and the relative difference in final cumulative dissipation must exceed 5%.

Current result. The lag difference is 27.174° and the relative dissipation difference is 78.74%, with the viscous run giving a mean lag of 2.689° and final cumulative dissipation 8.518119×10^{-5} , while the Maxwell-like run gives a mean lag of 29.863° and final cumulative dissipation 1.810825×10^{-5} .

f) 06_circularization: This case verifies the expected secular dissipative trend in an eccentric binary with active tides.

Pass criteria. Final eccentricity below the initial value, final semi-major axis below the initial value, and relative drift in the compensated energy diagnostic below 2%.

Current result. The orbit evolves from $e_0 = 0.4500$ to $e_f = 0.4375$ and from $a_0 = 2.6000$ to $a_f = 2.5478$, while the relative drift of mechanical plus dissipated energy is numerically zero to the reported precision.

g) 07_spin_synchronization: This case confirms that a super-synchronous spin is torqued toward the orbital frequency when a persistent tidal lag is present.

Pass criteria. The monitored spin ratio must decrease, the final value must be closer to 1 than the initial one, the mean bulge lag must exceed 0.2° , and the minimum spin ratio over the run must fall below the initial value.

Current result. The tracked ratio decreases from 4.0000 to 3.9820, the minimum value is 3.9820, and the mean bulge lag is 2.132° .

h) 08_obliquity_survey: This survey tests whether the model produces net obliquity damping across a small grid of tilted initial states and damping strengths rather than only in a single hand-picked example. The implementation surveys the grid $\{15^\circ, 45^\circ, 75^\circ\} \times \{0.10, 0.30, 0.60\}$ and measures the change in the spin-orbit angle over each run.

Pass criteria. At least half of the survey cases must have $\Delta\theta_{\text{spin-orbit}} < 0$, and the median $\Delta\theta_{\text{spin-orbit}}$ over the grid must also be negative.

Current result. The passing fraction is 0.667 and the median change is -0.0270° . Obliquity damping is therefore present across most, though not all, of the sampled parameter box.

B. Current status of the model

At present the suite reports *Passed 8 / 8 tests*. The nondissipative checks are particularly strong: the null-Kepler and conservative-equilibrium cases show only tiny residual drifts, at the level of 10^{-9} to 10^{-14} in the reported diagnostics. The quasi-static and rotational-flattening tests also indicate that the retained STF shape equations reproduce their intended instantaneous equilibria with substantial margin.

The dissipative tests show the expected qualitative behavior of the reduced model without spoiling the compensated energy bookkeeping. In the present run set the code clearly

distinguishes viscous and Maxwell-like responses, produces secular circularization and inspiral in eccentric systems, drives a super-synchronous spin in the correct direction, and yields net obliquity damping in a majority of the sampled survey cases. The last point should still be read as a trend rather than a theorem: the current survey indicates that the model is capable of obliquity damping over a useful region of parameter space, not that every admissible parameter choice must damp monotonically.

XII. CONCLUSION

A reduced Newtonian model of tidally deformable pseudo-fluid planets has been constructed in two layers. The first is a full modal framework with arbitrary angular modes, body-frame rotation, deformation-dependent spin inertia, a reduced rotational-flattening coupling implemented through the shape-dependent spin inertia, and generalized rheology. The second is a quadrupole implementation model suitable for direct numerical integration.

Relative to the simplest damped-quadrupole tide model, the present version is improved in four respects: it allows the spin inertia to vary with the deformation, it includes rotational flattening through the same reduced spin-shape coefficient that enters the spin kinetic energy, it supports frequency-dependent lag through internal relaxation branches, and it retains the leading angular momentum carried by the quadrupole mode itself. This last correction is small enough to preserve the basic ODE structure of the model, yet important enough to restore the explicit quadrupole torque in the spin balance and thereby improve the physical credibility of obliquity evolution.

The model remains a reduced model and not a full geophysical simulator. Nevertheless, it captures the key orbit-spin-shape couplings needed for tidal bulges, lag, synchronization, and orbital circularization. The standard one-tensor quadrupole closure also provides a mechanically consistent reduced description of spin evolution, including the leading modal angular momentum carried by the quadrupole. For applications in which secular obliquity damping is a primary target, however, the split-tensor extension (85)–(92) supplies the additional orientational degrees of freedom associated with distinct tidal and rotational bulges. The framework is therefore a useful intermediate model between point-mass celestial mechanics and full interior simulation.

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